Technical Document #370: Retrieval Augmented Generation - Comprehensive Overvi

Technical Document #370: Retrieval Augmented Generation - Comprehensive Overview

Hallucination detection identifies when LLMs generate factually incorrect information. It is critical for maintaining trust in RAG systems.

Chunking splits documents into smaller segments for embedding. Optimal chunk size balances context retention with retrieval precision.

Context window optimization selects the most relevant information to fit within the LLM's context limit. It maximizes information density.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Key Takeaways:

- Context window optimization selects the most relevant information to fit within the LLM's context limit.
- Vector databases store dense embeddings for semantic search.
- Chunking splits documents into smaller segments for embedding.